

HANNOVER, Germany

WMO: 103380

Lat: 52.4644N

Lon: 9.6781E

Elev: 59

StdP: 100.61

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.0	-6.8	-12.2	1.3	-7.3	-9.8	1.6	-5.4	12.6	7.6	11.3	7.2	3.8	90	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	30.2	19.7	28.1	19.0	26.1	18.1	20.7	27.7	19.8	26.2	19.0	24.6	4.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.7	22.9	17.9	12.9	21.9	17.0	12.2	21.1	59.9	27.3	56.7	26.5	54.1	24.6	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.9	8.0	DB	-12.0	34.1	3.4	1.6	-14.5	35.2	-16.5	36.1	-18.4	37.0	-20.9	38.2	(4)
(5)			WB	-12.3	23.0	3.5	0.8	-14.8	23.6	-16.9	24.0	-18.8	24.5	-21.3	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	1.7	2.2	5.3	10.1	13.3	16.8	19.1	18.2	15.2	10.9	6.5	3.8	(6)	
	DBStd	7.14	4.66	4.17	4.21	3.81	3.94	3.42	3.36	3.00	3.02	3.63	3.84	4.42	(7)	
	HDD10.0	1035	257	220	152	47	12	1	0	0	1	34	117	195	(8)	
	HDD18.3	3064	515	453	404	248	163	69	31	39	104	231	356	451	(9)	
	CDD10.0	1139	1	0	6	48	114	205	281	253	156	62	11	2	(10)	
	CDD18.3	128	0	0	0	7	24	54	34	9	0	0	0	0	(11)	
	CDH23.3	1217	0	0	0	9	78	229	524	292	81	4	0	0	(12)	
	CDH26.7	358	0	0	0	0	14	56	184	88	16	0	0	0	(13)	
(14) Wind	WSAvg	4.0	4.8	4.4	4.4	3.9	3.8	3.5	3.5	3.4	3.4	3.7	4.1	4.6	(14)	
(15) Precipitation	PrecAvg	642	52	38	45	43	59	63	65	67	52	48	53	58	(15)	
	PrecMax	938	150	111	133	107	171	156	158	172	120	136	114	160	(16)	
	PrecMin	433	3	4	10	7	8	20	22	9	6	11	2	6	(17)	
	PrecStd	110	29	22	26	24	35	31	31	30	26	26	25	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	13.9	18.9	24.8	28.8	31.0	33.9	32.8	29.0	23.1	16.9	14.0	(19)	
		MCWB	10.3	7.7	11.4	15.1	19.4	20.0	20.7	20.5	18.5	16.0	13.6	11.4	(20)	
	2%	DB	11.1	10.8	15.9	22.0	25.2	28.1	31.1	29.0	25.9	20.0	14.9	12.1	(21)	
		MCWB	9.2	8.3	10.4	13.3	17.2	19.1	19.5	19.7	17.4	15.6	12.5	10.3	(22)	
	5%	DB	9.2	9.2	13.2	19.9	23.1	26.0	28.2	26.2	23.1	17.9	13.2	10.9	(23)	
		MCWB	8.0	7.3	9.2	12.9	15.2	18.4	19.0	18.2	16.9	14.5	11.1	9.2	(24)	
	10%	DB	8.0	7.9	11.2	17.2	21.0	23.9	26.1	24.1	20.9	16.1	12.0	9.2	(25)	
		MCWB	6.9	6.2	8.1	11.9	14.9	17.2	18.4	17.2	16.2	13.6	10.0	7.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	9.8	12.8	15.8	20.0	21.6	22.2	21.6	19.7	16.8	14.5	11.7	(27)	
		MCDB	12.4	10.5	17.1	22.6	27.3	28.1	28.5	28.9	25.6	20.7	16.4	13.4	(28)	
	2%	WB	9.6	8.5	11.1	14.4	18.2	20.1	20.9	20.3	18.5	15.9	12.8	10.7	(29)	
		MCDB	11.0	10.2	14.7	20.0	23.2	25.9	27.9	27.1	24.1	19.6	14.4	12.1	(30)	
	5%	WB	8.1	7.3	10.0	13.4	16.7	18.9	20.0	19.3	17.5	14.9	11.5	9.4	(31)	
		MCDB	9.3	8.9	12.6	18.5	21.7	24.5	26.8	24.9	22.0	17.5	13.4	10.7	(32)	
	10%	WB	6.7	6.0	8.6	12.4	15.2	17.9	19.1	18.3	16.4	13.8	10.3	8.0	(33)	
		MCDB	7.6	7.5	11.2	17.1	19.7	22.8	24.9	23.0	20.1	16.2	11.9	9.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.4	7.6	10.1	10.0	10.0	9.9	9.5	8.9	7.0	4.9	4.4	(35)	
		MCDBR	5.0	7.3	11.4	14.4	13.6	13.7	14.0	13.4	12.9	9.3	6.1	5.3	(36)	
	5% WB	MCWBR	4.5	5.3	7.1	7.5	6.8	6.5	5.5	5.9	6.4	5.7	4.5	4.4	(37)	
		MCWBR	4.9	6.7	9.9	12.5	12.0	12.2	12.2	11.4	11.0	8.4	5.7	5.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.318	0.339	0.373	0.411	0.412	0.410	0.425	0.418	0.386	0.356	0.337	0.313	(40)	
		taud	2.372	2.357	2.279	2.218	2.252	2.276	2.243	2.278	2.360	2.426	2.401	2.363	(41)	
	Edn at Noon		672	768	808	821	840	844	821	805	788	739	642	606	(42)	
		Edh at Noon	60	80	105	126	128	127	129	118	97	75	58	51	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.64	1.33	2.39	3.94	4.83	5.17	4.93	4.14	3.02	1.73	0.78	0.50	(44)	
	RadStd		0.06	0.19	0.27	0.47	0.52	0.40	0.66	0.32	0.37	0.17	0.09	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-124	N/A	N/A	N/A

Nomenclature: See separate page